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CHAPTER 29 — COMPLETE MMU CHECKLIST & REFERENCE TABLES

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A maximum-detail, RTOS/QNX-style master checklist and consolidated reference tables covering every critical MMU concept, attribute, permission, barrier, descriptor field, and system-level rule.

This chapter is designed as a **single-page reference** you can use during debugging, design, or code reviews.

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29.1 — Master MMU Bring-Up Checklist

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Boot (EL3 → EL2 → EL1)

- Identity map boot code
- Identity map stack
- Identity map MMIO
- Configure MAIR
- Configure TCR
- Configure TTBR
- DSB SY
- ISB
- Enable MMU
- Enable caches
- Enable SMP

EL3

- Secure/non-secure split
- Secure MMIO mapping
- Secure RAM mapping

EL2

- Build Stage-2 tables
- Configure HCR_EL2

- Configure VTTBR_EL2
- Map GIC MMIO
- Map timer MMIO
- Enable Stage-2 MMU

EL1

- Build TTBR1 (kernel)
- Build TTBR0 (user)
- Map vectors
- Map kernel code/data
- Map MMIO
- Enable Stage-1 MMU

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29.2 — Master Descriptor Checklist

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Descriptor Fields

- Type: block / page / table
- PA base
- AP (permissions)
- PXN/UXN (execute control)
- SH (shareability)
- AttrIndx (MAIR index)
- AF (access flag)
- Contiguous bit

- Valid bit

Correct Values

- Kernel code → RO + executable
- Kernel data → RW + non-exec
- User code → RO + executable
- User data → RW + non-exec
- MMIO → RW + non-exec + Device-nGnRnE
- DMA buffers → Normal WB or NC

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29.3 — Master MAIR Table

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AttrIndx	Type	Meaning
0x00	Device-nGnRnE	Strongly ordered MMIO
0x04	Device-nGnRE	Early-ack device
0x08	Device-nGRE	Limited reordering
0x44	Normal NC	Non-cacheable RAM
0xBB	Normal WT	Write-through RAM
0xFF	Normal WB	Write-back RAM (best performance)

Rules

- MMIO → 0x00
- RAM → 0xFF
- DMA buffers → 0xFF or 0x44

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29.4 — Master TCR Checklist

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Granule

- TG0/TG1 = 4 KB (most common)
- TG0/TG1 = 16 KB or 64 KB (rare)

Shareability

- SH0/SH1 = Inner Shareable

Cacheability

- IRGN0/IRGN1 = WB
- ORGN0/ORGN1 = WB

Address Size

- IPS = physical address size
- T0SZ/T1SZ = VA size

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29.5 — Master TTBR Checklist

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TTBR0_EL1

- user space
- ASID-tagged
- per-process mappings

TTBR1_EL1

- kernel space
- global mappings
- blocks preferred

TTBR0_EL2

- Stage-2 tables
- VMID-tagged

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29.6 — Master Permission Table

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AP	Meaning
00	Kernel RW, User RW
01	Kernel RW, User RO
10	Kernel RW, User NO
11	Kernel RO, User NO

Execute Control

- PXN = privileged execute never
- UXN = user execute never

Rules

- code → executable
- data → non-exec
- MMIO → non-exec

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29.7 — Master Shareability Table

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SH	Meaning
00	Non-Shareable
10	Outer Shareable
11	Inner Shareable

Rules

- RAM → Inner Shareable
- DMA buffers → Inner Shareable
- MMIO → Non-Shareable

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29.8 — Master Barrier Checklist

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DSB SY

- complete all memory operations
- required after MMU changes
- required after descriptor writes
- required after MMIO writes

DMB SY

- ordering only
- rarely used for MMU

ISB

- flush pipeline
- required after MMU enable
- required after TLBI

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29.9 — Master TLBI Checklist

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Correct TLBI Sequence

```
DSB SY
TLBI <operation>
DSB SY
ISB
```

Common TLBI Operations

- TLBI VAE1 → invalidate VA at EL1
- TLBI ALLE1 → invalidate all EL1
- TLBI VAE2 → invalidate VA at EL2
- TLBI ALLE2 → invalidate all EL2

Rules

- always follow TLBI with DSB SY + ISB
- always use BBM for descriptor replacement

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29.10 — Master Cache Maintenance Checklist

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Before DMA Read

```
DC CVAC
DSB SY
```

After DMA Write

```
DC CIVAC
DSB SY
```

Instruction Cache

```
DC CVAC
DSB SY
IC IVAU
DSB SY
ISB
```

Rules

- never clean MMIO

- never invalidate MMIO

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29.11 — Master Device Mapping Checklist

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MMIO

- AttrIndx = 2 (Device-nGnRnE)
- SH = 00 (Non-Shareable)
- AP = RW
- PXN = 1
- UXN = 1
- 4 KB pages only

Ordering

```
STR → DSB SY
DSB SY → LDR
```

Never

- map MMIO as Normal
- use blocks for MMIO
- allow speculation

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29.12 — Master Stage-2 Checklist

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Stage-2 Descriptor Fields

- S2AP
- S2PXN/S2UXN
- S2AttrIndx
- S2SH
- IPA base

Rules

- guest RAM → blocks
- guest MMIO → pages
- hypervisor overrides guest attributes
- VMID required for isolation

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29.13 — Master Security Checklist

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W^X

- code → RO + exec
- data → RW + non-exec

PAN

- prevent EL0 from accessing EL1

PXN/UXN

- prevent execution from data

WXN

- global write-xor-execute

TrustZone

- secure vs non-secure split

PAC/BTI

- protect control flow

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29.14 — Master SMP Checklist

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Requirements

- Inner Shareable RAM
- broadcast TLBI
- broadcast cache maintenance
- identical TTBR1_EL1 on all cores
- identical TCR/MAIR

Never

- map RAM as Non-Shareable

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29.15 — Master Debugging Checklist

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Dump

- descriptors
- TLB
- walk-cache
- ESR/FAR/HPFAR
- Stage-2 tables

Trace

- barriers
- cache maintenance
- MMIO
- DMA
- interrupts

Check

- alignment
- granule
- attributes
- permissions
- shareability

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29.16 — Master Failure Mode Table

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Failure	Cause	Fix
Device malfunction	MMIO mapped as Normal	Use Device-nGnRnE
SMP incoherence	RAM Non-Shareable	Use Inner Shareable
Page faults	wrong AP/PXN/UXN	fix permissions
DMA corruption	wrong attributes	WB/NC + cache ops
Stale TLB	missing TLBI	full TLBI sequence
Stale walk-cache	missing BBM	BBM sequence
Boot crash	missing identity map	identity map early memory
Interrupt failure	GIC mapped wrong	Device-nGnRnE

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29.17 — Chapter Summary

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This chapter provided:

- the **complete MMU checklist**
- consolidated **reference tables**
- descriptor rules
- MAIR/TCR/TTBR rules
- permission rules
- shareability rules
- barrier rules
- TLBI rules
- cache maintenance rules
- device mapping rules
- Stage-2 rules
- security rules
- SMP rules
- debugging rules
- failure mode tables